MATHEMATICS II FOR DATA SCIENCE
GA-4 SOLUTIONS MAY 2024

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1. Question

Which of the following statements are correct?

- ☐ If A and B are two square matrices, then Rank(AB)=Rank(BA).
- ☐ If A and B are two 2×2 square matrices, then Rank$(AB) \leq$ Rank(B).
- ☐ If A and B are two 2×2 square matrices, then Rank$(AB) \leq$ Rank(A).
- ☐ If A and B are $n \times n$ matrices of rank n, then AB has rank n.

1.1. Answers:

- If A and B are two 2×2 square matrices, then Rank$(AB) \leq$ Rank(B).
- If A and B are two 2×2 square matrices, then Rank$(AB) \leq$ Rank(A).
- If A and B are $n \times n$ matrices of rank n, then AB has rank n.

1.2. Solution

Few observations:

- Every row in matrix AB is a linear combination of the rows of matrix B.
- Every column in matrix AB is a linear combination of the columns of matrix A.
- Row rank of a matrix is equal to its column rank.

Let $A = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$ and $B = \begin{bmatrix} w & x \\ y & z \end{bmatrix}$. $AB = \begin{bmatrix} pw + qy & px + qz \\ rw + sy & rx + sz \end{bmatrix}$. Notice that the first row is

$p(w, x) + q(y, z)$ and the second row is $r(w, x) + s(y, z)$. The first column is $w(p, r) + y(q, s)$ and the second column is $x(p, r) + z(q, s)$.

From the above observations, we can see that the rows of AB is contained in the row space of B and the columns of AB is contained in the columns of A. Therefore, $Rank(AB) \leqslant Rank(A)$.
$Row\ Rank(AB) \leqslant Row\ Rank(B)$. Since row rank of a matrix is equal to its column rank, we can say that $Rank(AB) \leqslant Rank(B)$. Therefore, **option 2** and **option 3** are correct.

If A and B have are $n \times n$ matrices of rank n, then $Rank(AB) = n$. O**ption 4** is correct.

If a square matrix has full rank, then the determinant of that matrix is not equal to zero. Therefore, $det(A) \neq 0$ and $det(B) \neq 0$.

$$det(AB) = det(A) \cdot det(B) \neq 0$$

Therefore, AB has full rank, which is n.

Option 1 is incorrect. Let us take $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} and\ B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. $AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} and\ BA = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. We can clearly see that $Rank(AB) = 1$ but $Rank(BA) = 0$.

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2. Question

Match the vector spaces (with the usual scalar multiplication and vector addition as in $M_{3\times3}(\mathbb{R})$ in column A with their bases in column B and the dimensions of the vector spaces in column C in $Table\ :\ M2W4GA1$.

	Vector space (Column A)		Basis (Column B)		Dimension of the vector space (Column C)
a)	$V = \left\{ \begin{bmatrix} x & y & z \\ 0 & z & x \\ y & 0 & 0 \end{bmatrix} \mid x + 2y = z, \text{ and } x, y, z \in \mathbb{R} \right\}$	i)	$\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$	1)	6
b)	$V = \{A \mid A \in M_{3\times3}(\mathbb{R}),\ A \text{ is a symmetric matrix } \}$	ii)	$\left\{ \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 2 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{bmatrix} \right\}$	2)	6
c)	$V = \{A \mid A \in M_{3\times3}(\mathbb{R}),\ A \text{ is an upper triangular matrix } \}$	iii)	$\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$	3)	2

Table : M2W4GA1

Choose the correct option.

- [ ] a $\rightarrow$ i $\rightarrow$ 2.
- [ ] b $\rightarrow$ i $\rightarrow$ 2.
- [ ] a $\rightarrow$ ii $\rightarrow$ 3.
- [ ] c $\rightarrow$ iii $\rightarrow$ 1
- [ ] b $\rightarrow$ iii $\rightarrow$ 2.
- [ ] c $\rightarrow$ i $\rightarrow$ 1.

2.1. Answers :

- a $\rightarrow$ ii $\rightarrow$ 3.
- b $\rightarrow$ iii $\rightarrow$ 2.
- c $\rightarrow$ i $\rightarrow$ 1.

2.2. Solution

Option 3 is correct and **option 1** is incorrect.

$$V = \left\{ \begin{bmatrix} x & y & z \\ 0 & z & x \\ y & 0 & 0 \end{bmatrix} \mid x + 2y = z \; and \; x, y, z \in \mathbb{R} \right\}$$

$$V = \left\{ \begin{bmatrix} x & y & x+2y \\ 0 & x+2y & x \\ y & 0 & 0 \end{bmatrix} \mid x, y, z \in \mathbb{R} \right\}$$

$$V = \left\{ x \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} + y \begin{bmatrix} 0 & 1 & 2 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{bmatrix} \mid x, y, z \in \mathbb{R} \right\}$$

$$V = span \left\{ \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 2 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{bmatrix} \right\}$$

Option 5 is correct and **option 2** is incorrect.

$$V = \{A | A \in M_{3\times3}(\mathbb{R}).\ \ A \textit{ is a symmetric matrix}.$$

$$V = \left\{ \begin{bmatrix} x & y & z \\ y & p & q \\ z & q & r \end{bmatrix} \mid x, y, z, p, q, r \in \mathbb{R} \right\}$$

$$V = \left\{ \begin{bmatrix} x & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & y & 0 \\ y & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & z \\ 0 & 0 & 0 \\ z & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & p & 0 \\ 0 & 0 & 0 \end{bmatrix} \right.$$
$$\left. + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & q \\ 0 & q & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & r \end{bmatrix} \mid x, y, z, p, q, r \in \mathbb{R} \right\}$$

$$V = \left\{ x \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + y \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + z \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} + p \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \right.$$
$$\left. q \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} + r \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mid x, y, z, p, q, r \in \mathbb{R} \right\}$$

$$V = span \left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \right.$$
$$\left. \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$$

Option 6 is correct and **option 4** is incorrect.

$$V = \{A | A \in M_{3\times3}(\mathbb{R}).\ \ A \textit{ is an upper triangular matrix}.$$

$$V = \left\{ \begin{bmatrix} x & y & z \\ 0 & p & q \\ 0 & 0 & r \end{bmatrix} \mid x, y, z, p, q, r \in \mathbb{R} \right\}$$

$$V = \left\{ \begin{bmatrix} x & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & y & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & z \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & p & 0 \\ 0 & 0 & 0 \end{bmatrix} \right.$$
$$\left. + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & q \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & r \end{bmatrix} \mid x, y, z, p, q, r \in \mathbb{R} \right\}$$

$$V = \left\{ x\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + y\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + z\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + p\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + q\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} + r\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mid x, y, z, p, q, r \in \mathbb{R} \right\}$$

$$V = span\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$$

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3. Question

Find the dimension of the vector space

$$V = \{(x, y, z) \mid x, y, z \in \mathbb{R}, x + y + z = 0, z = 0\}$$
$$Addition: (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2);$$
$$(x_1, y_1, z_1), (x_2, y_2, z_2) \in V$$
$$Scalar\ multiplication: c(x, y, z) = (cx, cy, cz);\ (x, y, z) \in V,\ c \in \mathbb{R}$$

3.1. Answer : 1

3.2. Solution

Every vector in V is a solution for $x + y + z = 0$ and $z = 0$. The matrix form $Ax = b$ for these two equations is given below.

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix},\ x = \begin{bmatrix} x \\ y \\ z \end{bmatrix},\ b = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Let us augment A and B.

$$(A|b) = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

The reduced row echelon form of $(A|b)$ can be obtained by subtracting the second row from the first row:

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

We can see that there is one independent variable $\implies z$. On solving the above equations, we get

$x = -y \; and \; z = 0$

The solution vector is of the form $\begin{bmatrix} -y \\ y \\ 0 \end{bmatrix} \implies y \cdot \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$.

$\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ is a basis vector for the vector space V. Hence, the dimension of the vector space V is 1.

The intersection of the planes $x + y + z = 0$ and $z = 0$ will be a straight line, which also suggests that the dimension of V will be 1.

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4. Question

Find the rank of the matrix A, where $A = [a_{ij}]$ is of order 2021×2021 and $a_{ij} = min\{i, j\}, i, j = 1, 2, \ldots 2021$. [Hint: First do it for smaller matrices, such as 3×3 or 4×4.]

4.1. Answer : 2021

4.2. Solution

It is given that every $ij^{th} \; element \; will \; be \; min\{i, j\}$, where i refers to the row number and j refers to the column number .Let us look at a 4×4 matrix as an example.

$$P = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{bmatrix}$$

Let us apply elementary row operations on P.

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{bmatrix} \xrightarrow{R2-R1} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{bmatrix} \xrightarrow{R3-R1} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 1 & 2 & 3 & 4 \end{bmatrix} \xrightarrow{R4-R1} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & 3 \end{bmatrix}$$

$$\xrightarrow{R3-R2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 2 & 3 \end{bmatrix} \xrightarrow{R4-R2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \xrightarrow{R4-R3} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The above matrix is an upper triangular matrix. It is in row echelon form. The product of the diagonals of an upper triangular matrix is the determinant of that matrix. Therefore, the determinant of the above matrix is 1.

The only row operation performed above was:- add the multiple of a row to another row. This operation does not change the determinant of the original matrix. therefore $det(P) = 1$.

If we take the matrix given in the question and convert it into its row echelon form using elementary row

operations, we will once again get an upper triangular matrix with all the non-zero entries as 1. Therefore, the determinant of the matrix given in the question is 1. If the determinant of a square matrix of order $n \times n$ is not equal to zero, then the rank of that matrix is n.

Therefore, the rank of the given matrix is 2021

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5. Question

Consider a subspace $W = \{(x_1, x_2, x_3, x_4, x_5) |\ 8x_1 + 2x_5 = 0\ and\ 4x_2 + 5x_4 = 0\}$ of $\mathbb{R}^5$.The dimension of
W is

5.1. Answer : 3

5.2. Solution

Every vector in W is a solution for $8x_1 + 2x_5 = 0\ and\ 4x_2 + 5x_4 = 0$. The matrix form $Ax = b$ for these two equations is given below.

$$A = \begin{bmatrix} 8 & 0 & 0 & 0 & 2 \\ 0 & 4 & 0 & 5 & 0 \end{bmatrix},\ x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix},\ b = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Let us augment A and b.

$$(A|b) = \left[\begin{array}{ccccc|c} 8 & 0 & 0 & 0 & 2 & 0 \\ 0 & 4 & 0 & 5 & 0 & 0 \end{array}\right]$$

Let us apply row operations of $(A|b)$.

$$\left[\begin{array}{ccccc|c} 8 & 0 & 0 & 0 & 2 & 0 \\ 0 & 4 & 0 & 5 & 0 & 0 \end{array}\right] \xrightarrow{\frac{R1}{8}} \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & 0.25 & 0 \\ 0 & 4 & 0 & 5 & 0 & 0 \end{array}\right] \xrightarrow{\frac{R2}{4}} \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & 0.25 & 0 \\ 0 & 1 & 0 & 1.25 & 0 & 0 \end{array}\right]$$

The above matrix is in reduced row echelon form. We can see that there are three independent variables $\implies x_3, x_4, x_5$. From the matrix above, we get the following equations.

$$x_1 + 0.25x_5 = 0 \implies x1 = -0.25x_5$$
$$x_2 + 1.25x_4 = 0 \implies x_2 = -1.25x_4$$

The solution vector is of the form $\begin{bmatrix} -0.25x_5 \\ -1.25x_4 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \implies x_3 \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \cdot \begin{bmatrix} 0 \\ -1.25 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_5 \cdot \begin{bmatrix} -0.25 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$.

We can see that the basis set for the subspace W will have 3 vectors. Therefore, the dimension of W is 3.

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6. Question

Consider a matrix $A = \begin{bmatrix} 2 & -4 & 1 & 1 \\ 1 & -3 & 1 & -4 \\ 3 & -7 & 2 & 3 \end{bmatrix}$. The rank of the matrix is

6.1. Answer : 3

6.2. Solution

Let us apply elementary row operations on matrix A.

$$\begin{bmatrix} 2 & -4 & 1 & 1 \\ 1 & -3 & 1 & -4 \\ 3 & -7 & 2 & 3 \end{bmatrix} \xrightarrow{R3-R1} \begin{bmatrix} 2 & -4 & 1 & 1 \\ 1 & -3 & 1 & -4 \\ 1 & -3 & 1 & 2 \end{bmatrix} \xrightarrow{R3-R2} \begin{bmatrix} 2 & -4 & 1 & 1 \\ 1 & -3 & 1 & -4 \\ 0 & 0 & 0 & 6 \end{bmatrix} \xrightarrow{R1 \rightleftarrows R2} \begin{bmatrix} 1 & -3 & 1 & -4 \\ 2 & -4 & 1 & 1 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

$$\xrightarrow{\frac{R3}{6}} \begin{bmatrix} 1 & -3 & 1 & -4 \\ 2 & -4 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\frac{R2}{2}} \begin{bmatrix} 1 & -3 & 1 & -4 \\ 1 & -2 & 1/2 & 1/2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R2-R1} \begin{bmatrix} 1 & -3 & 1 & -4 \\ 0 & 1 & -1/2 & 9/2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The above matrix is in row echelon form. We can see that there are three columns with leading entries. So the basis set of the column space of matrix A will have 3 vectors. Therefore, the rank matrix A in 3.

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7. Question

Consider two matrices $A = \begin{bmatrix} -1 & -6 & 6 \\ -10 & 4 & 2 \\ -8 & 7 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & -10 & -8 \\ -6 & 4 & 7 \\ 6 & 2 & -4 \end{bmatrix}$. Then $Rank(A) - Rank(B)$ is

7.1. Answer : 0

7.2. Solution

Let us apply row operations on matrix A.

$$\begin{bmatrix} -1 & -6 & 6 \\ -10 & 4 & 2 \\ -8 & 7 & -4 \end{bmatrix} \xrightarrow{(-1)\cdot R1} \begin{bmatrix} 1 & 6 & -6 \\ -10 & 4 & 2 \\ -8 & 7 & -4 \end{bmatrix} \xrightarrow{R2+10\cdot R1} \begin{bmatrix} 1 & 6 & -6 \\ 0 & 64 & -58 \\ -8 & 7 & -4 \end{bmatrix} \xrightarrow{R3+8R1} \begin{bmatrix} 1 & 6 & -6 \\ 0 & 64 & -58 \\ 0 & 55 & -52 \end{bmatrix}$$

$$\xrightarrow{R3-\left(\frac{55}{64}\right)R2} \begin{bmatrix} 1 & 6 & -6 \\ 0 & 64 & -58 \\ 0 & 0 & -52 + \left(\frac{55 \times 58}{64}\right) \end{bmatrix} \xrightarrow{\frac{R2}{64}} \begin{bmatrix} 1 & 6 & -6 \\ 0 & 1 & -58/64 \\ 0 & 0 & -138/64 \end{bmatrix} \xrightarrow{\left(\frac{-64}{136}\right)R3} \begin{bmatrix} 1 & 6 & -6 \\ 0 & 1 & -58/64 \\ 0 & 0 & 1 \end{bmatrix}$$

The above matrix is in row echelon form. There are three columns with leading entries. Therefore, $Rank(A) = 3$

Let us apply row operations on matrix B.

$$\begin{bmatrix} -1 & -10 & -8 \\ -6 & 4 & 7 \\ 6 & 2 & -4 \end{bmatrix} \xrightarrow{(-1)\cdot R1} \begin{bmatrix} 1 & 10 & 8 \\ -6 & 4 & 7 \\ 6 & 2 & -4 \end{bmatrix} \xrightarrow{R2+6\cdot R1} \begin{bmatrix} 1 & 10 & 8 \\ 0 & 64 & 55 \\ 6 & 2 & -4 \end{bmatrix} \xrightarrow{R3-6R1} \begin{bmatrix} 1 & 10 & 8 \\ 0 & 64 & 63 \\ 0 & -58 & -52 \end{bmatrix}$$

$$\xrightarrow{R3+\left(\frac{58}{64}\right)R2} \begin{bmatrix} 1 & 10 & 8 \\ 0 & 64 & 63 \\ 0 & 0 & -52+\left(\frac{63\times 58}{64}\right) \end{bmatrix} \xrightarrow{\frac{R2}{64}} \begin{bmatrix} 1 & 10 & 8 \\ 0 & 1 & 63/64 \\ 0 & 0 & 326/64 \end{bmatrix} \xrightarrow{\left(\frac{64}{326}\right)R3} \begin{bmatrix} 1 & 10 & 8 \\ 0 & 1 & 63/64 \\ 0 & 0 & 1 \end{bmatrix}$$

The above matrix is in row echelon form. There are three columns with leading entries. Therefore, $Rank(B) = 3$

$$Rank(A) - Rank(B) \implies 3 - 3 \implies 0$$

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8. Question

If V is the subspace spanned by these two vectors $(-1, 0, 1)$ and $(2, 1, -1)$ in $\mathbb{R}^3$, then how many of the following vectors will be in V? $(1, 1, 0)$, $(5, 2, -3)$, $(8, 5, -4)$, $(-3, 0, 3)$, $\left(\frac{-5}{6}, \frac{5}{6}, \frac{5}{6}\right)$, $(-2, 3, 3)$.

8.1. Answer : 3

8.2. Solution

Let us check if $(-1, 0, 1)$ and $(2, 1, -1)$ are linearly independent or not.

$$p \cdot (-1, 0, 1) + q \cdot (2, 1, -1) = (0, 0, 0) \implies (2q - p, q, p - q) = (0, 0, 0)$$

We get the following equations:

$$\begin{aligned} 2q - p &= 0 \quad (1) \\ q &= 0 \quad (2) \\ p - q &= 0 \quad (3) \end{aligned}$$

From (2), we can see that $q = 0$. Substitute $q = 0$ in (3) to get $p = 0$. $p = 0$ *and* $q = 0$ is the only solution for $p \cdot (-1, 0, 1) + q \cdot (2, 1, -1) = (0, 0, 0)$. Therefore, $(-1, 0, 1)$ and $(2, 1, -1)$ are linearly independent.

Let us have five matrices A, B, C, D, E *and* F. The first two rows of these six matrices will have $(-1, 0, 1)$ and $(2, 1, -1)$ respectively. The last row of A will have $(1, 1, 0)$, the last row of B will have $(5, 2, -3)$, the last row of C will have $(8, 5, -4)$, the last row of D will have $(-3, 0, 3)$, the last row of E will have $\left(\frac{-5}{6}, \frac{5}{6}, \frac{5}{6}\right)$ and the last row of F will have $(-2, 3, 3)$.

$$A = \begin{bmatrix} -1 & 0 & 1 \\ 2 & 1 & -1 \\ 1 & 1 & 0 \end{bmatrix}, B = \begin{bmatrix} -1 & 0 & 1 \\ 2 & 1 & -1 \\ 5 & 2 & -3 \end{bmatrix}, C = \begin{bmatrix} -1 & 0 & 1 \\ 2 & 1 & -1 \\ 8 & 5 & -4 \end{bmatrix},$$

$$D = \begin{bmatrix} -1 & 0 & 1 \\ 2 & 1 & -1 \\ -3 & 0 & 3 \end{bmatrix}, E = \begin{bmatrix} -1 & 0 & 1 \\ 2 & 1 & -1 \\ -(5/6) & (5/6) & (5/6) \end{bmatrix}, F = \begin{bmatrix} -1 & 0 & 1 \\ 2 & 1 & -1 \\ -2 & 3 & 3 \end{bmatrix}$$

Let us calculate the determinant for these six matrices.

$det(A)$	$= -1(0-(-1)) + 1(2-1)$ $= -1(1) + 1(1)$ $= 0$
$det(B)$	$= -1(-3-(-2)) + 1(4-5)$ $= -1(-1) + 1(-1)$ $= 0$
$det(C)$	$= -1(-4-(-5)) + 1(10-8)$ $= -1(1) + 1(2)$ $= 1$
$det(D)$	$= -1(3-0) + 1(0-(-3))$ $= -3 + 3$ $= 0$
$det(E)$	$= -1((5/6)-(-(5/6)) + 1((10/6)-(-(5/6)))$ $= -1(10/6) + 1(15/6)$ $= (5/6)$
$det(F)$	$= -1(3-(-3)) + 1(6-(-2))$ $= -1(6) + 1(8)$ $= 2$

The rows of a matrix are linearly independent if the determinant of that matrix is not zero. From the above table, we can see that the rows of matrix A, B and matrix D are not linearly independent. Remember, the first two rows are linearly independent. From this we can infer that $(1,1,0)$,$(5,2,-3)$ and $(-3,0,3)$ can be expressed as a linear combination of $(-1,0,1)$ and $(2,1,-1)$. Therefore, $(1,1,0)$, $(5,2,-3)$ and $(-3,0,3)$ belong to V.

The other three vectors do not belong to V as they are linearly independent of $(-1,0,1)$ and $(2,1,-1)$.

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9. Question

Find the value of c for which the vector $(15,26,c)$ will be in the spanning set of the vectors $(1,0,1)$ and $(0,1,-1)$ in R^3 with usual addition and scalar multiplication.

9.1. Answer : –11

9.2. Solution

Let rows of matrix A have the vectors given in the question. $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 15 & 26 & c \end{bmatrix}$. If $det(A)$ is equal to zero, then the rows of matrix A are not linearly independent. This would mean that every row in matix A can be expressed as a linear combination of the remaining rows.

$$\begin{aligned} det(A) &= 1(c-(-26)) + 1(0-15) \\ det(A) &= c + 26 - 15 \\ det(A) &= c + 11 \end{aligned}$$

Let us solve the following equality.

$$\begin{aligned} det(A) &= 0 \\ c + 11 &= 0 \\ c &= -11 \end{aligned}$$

If $c = -11$, then $(15, 26.c)$ will be in $span\{(1,0,1),(0,1,-1)\}$.

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10. Question

Consider the subspace $V = \{(x,y,x)|z = 7x + 2y,\ where\ x,y,z \in \mathbb{R}\}$ of $\mathbb{R}^3$ and $span(\{(p,0,1),(0,q,1)\})$ = V. Find the value of $\frac{1}{pq}$.

10.1. Answer : 14

10.2. Solution

If $V = span(\{(p,0,1),(0,q,1)\})$, then $(p,0,1)$ and $(0,q,1)$ will satisfy $z = 7x + 2y$.

$$\begin{aligned} z = 7x + 2y \implies 1 = 7(p) + 2(0) \implies 1 = 7p \implies p = 1/7 \\ z = 7x + 2y \implies 1 = 7(0) + 2(q) \implies 1 = 2q \implies q = 1/2 \end{aligned}$$

$$\frac{1}{pq} \implies \frac{1}{\left(\frac{1}{7}\right)\left(\frac{1}{2}\right)} \implies \frac{1}{\frac{1}{14}} \implies 14$$

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11. Question

Suppose V is a vector space defined as $V = \{A \mid A \in M_{4\times4}(\mathbb{R}),$ A is an upper triangular matrix, and the sum of the diagonal entries is zero$\}$. What is the cardinality of a basis of V?

11.1. Answer : 9

11.2. Solution

Every vector in V will be of the form $\begin{bmatrix} a_1 & a_2 & a_3 & a4 \\ 0 & a_5 & a_6 & a_7 \\ 0 & 0 & a_8 & a_9 \\ 0 & 0 & 0 & a_{10} \end{bmatrix}$, where $a_i \in \mathbb{R}$, for all $i\ in\ \{1, 2, \cdots, 10\}$.

It is also given that $a_1 + a_5 + a_8, a_{10} = 0 \implies a_{10} = -a_1 - a_5 - a_8$.

$$\begin{bmatrix} a_1 & a_2 & a_3 & a4 \\ 0 & a_5 & a_6 & a_7 \\ 0 & 0 & a_8 & a_9 \\ 0 & 0 & 0 & a_{10} \end{bmatrix} \implies \begin{bmatrix} a_1 & a_2 & a_3 & a4 \\ 0 & a_5 & a_6 & a_7 \\ 0 & 0 & a_8 & a_9 \\ 0 & 0 & 0 & -a_1 - a_5 - a_8 \end{bmatrix}$$

The above matrix can be expressed as:

$$a_1 \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} + a_2 \cdot \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + a_3 \cdot \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + a_4 \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + a_5 \cdot \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

$$a_6 \cdot \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + a_7 \cdot \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + a_8 \cdot \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} + a_9 \cdot \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Every vector in V is a linear combination of the nine vectors above. Therefore, the nine vectors above will form the basis set for vector space V.

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12. Question

U is the vector space of all upper triangular matrices of order 3. V is the vector space of all lower triangular matrices of order 3. The dimensions of the vector spaces $U, V, U \cap V$ are a, b, c respectively. Find $a + b + c$. The usual rules of addition and scalar multiplication apply for all three spaces.

12.1. Answer : 15

12.2. Solution

Every vector in U is a linear combination of the matrices below. The set below forms the basis for U.

$$\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right.$$

Every vector in V is a linear combination of the matrices below. The set below forms the basis for V.

$$\left\{\begin{bmatrix}1&0&0\\0&0&0\\0&0&0\end{bmatrix},\begin{bmatrix}0&1&0\\0&0&0\\0&0&0\end{bmatrix},\begin{bmatrix}0&0&1\\0&0&0\\0&0&0\end{bmatrix},\begin{bmatrix}0&0&0\\0&1&0\\0&0&0\end{bmatrix},\begin{bmatrix}0&0&0\\0&0&1\\0&0&0\end{bmatrix},\begin{bmatrix}0&0&0\\0&0&0\\0&0&1\end{bmatrix}\right.$$

The dimension of both the vector spaces U and V is 6. Therefore, $a = 6$ and $b = 6$.

The intersection of U and V will be the set of all diagonal matrices. This set is also a vector space. Every vector in $U \cap V$ is a linear combination of the matrices below. The set below forms the basis for $U \cap V$.

$$\left\{\begin{bmatrix}1&0&0\\0&0&0\\0&0&0\end{bmatrix},\begin{bmatrix}0&0&0\\0&1&0\\0&0&0\end{bmatrix},\begin{bmatrix}0&0&0\\0&0&0\\0&0&1\end{bmatrix}\right\}$$

The dimension of $U \cap V$ is 3. Therefore, $c = 3$.

$$a + b + c \Longrightarrow 6 + 6 + 3 \Longrightarrow 15$$